

Quantum Computing

Part 5 - Hardware by Modality: Spin Qubits

Compiled by the Spaced Research Ledger

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A spin qubit stores information in the magnetic orientation of a single electron, held inside a silicon structure called a quantum dot. The qubits are roughly a hundred nanometres across, which makes them thousands of times smaller than the superconducting circuits in Part 1.

Size is not the real argument for them. The argument is the factory. These devices are made on the same 300-millimetre wafer lines, with the same lithography, that produce ordinary computer chips. No other modality in this collection can be manufactured by an existing semiconductor industry without inventing a new process for it.

Two words recur below and are worth settling now. CMOS is the standard manufacturing process behind nearly all commercial chips. EUV, or extreme ultraviolet lithography, is the light source that patterns the finest features on those chips. When a result in this part cites both, the claim is that quantum devices came off a production line rather than a laboratory bench.

Section 1 covers Intel, Section 2 Diraq, Section 3 imec, and Section 4 Quantum Motion. Section 5 covers germanium, a competing material. Sections 6 and 7 cover the two groups holding the fidelity records, at UNSW and RIKEN.

1. Intel

Intel's contribution is manufacturing data rather than machines. It measures spin qubits across whole wafers using its own production tooling, which produces the kind of yield and uniformity statistics no laboratory can generate.

Recent Developments

Intel demonstrated tune-up of a 12-quantum-dot spin-qubit device, known as Tunnel Falls [1]. It is the largest reported qubit count in a single spin-qubit device fabricated on a 300-millimetre semiconductor line using extreme ultraviolet and high-volume manufacturing processes.

Current State

Intel published a Nature paper in May 2024 demonstrating uniformity, fidelity and measurement statistics for CMOS-fabricated spin qubits, using a 300-millimetre cryogenic wafer-probing process [2]. Single-electron devices from those wafers achieved 99.9% gate fidelity, the highest reported for qubits made entirely with CMOS industry manufacturing [2].

The fabrication details carry the argument. Devices were made using extreme ultraviolet lithography on 300-millimetre wafers, with qubit feature sizes of about 100 nanometres, while remaining compatible with high-volume CMOS manufacturing [2].

The probing process is the quieter contribution. Intel's high-throughput cryogenic probing generated far more device data than earlier methods, allowing analysis of qubit uniformity and yield across whole wafers [2]. Knowing what fraction of a wafer works is a manufacturing question, and nobody asks it of the machines in Parts 2 through 4.

2. Diraq

Diraq's results answer the question that matters most for this modality. Its high fidelities come from randomly chosen devices at more than one foundry, which makes them a statement about a process rather than about a chip.

Recent Developments

Diraq's eight-qubit silicon spin-qubit array, fabricated on a 300-millimetre silicon MOS foundry process, was published in Nature Communications [3].

Current State

The reproducibility result is the important one. Diraq-designed, imec-fabricated two-qubit devices, made with standard 300-millimetre foundry tooling, achieved single- and two-qubit control fidelities above 99% for all four tested devices [4]. State preparation and measurement fidelities reached 99.9%, confirmed by gate set tomography in a peer-reviewed Nature paper.

Those devices were randomly selected rather than curated [5]. Picking the best chip from a batch is normal practice in research and tells you nothing about yield. Measuring randomly chosen devices demonstrates that high fidelity is reproducible across foundry-fabricated silicon spin qubits.

Temperature is the other advantage. Diraq reported a hot-qubit result of 99.85% single-qubit fidelity and 98.92% two-qubit fidelity operating at 1 kelvin [6]. That is roughly ten times warmer than the 15-millikelvin environment superconducting platforms need. Cooling to 1 kelvin is enormously cheaper, and it frees room in the refrigerator for control electronics.

Diraq is a Sydney-based spinout from the University of New South Wales, founded by Andrew Dzurak and Tuomo Tanntu in 2022 [6]. It builds on standard CMOS processes at both GlobalFoundries and imec, and has demonstrated high-fidelity qubits on 300-millimetre wafers at both [6]. Working at two foundries indicates the process is not tied to one supplier.

3. imec

imec is the shared research foundry behind several results in this part, including Diraq's. Its own contribution is lithography, and it has now put quantum dots through the most advanced patterning tools the chip industry has.

Recent Developments

imec presented the first quantum dot qubit device fabricated using High NA extreme ultraviolet lithography [8]. It patterned gaps as small as 6 nanometres between plunger and barrier gates on a working qubit array. High NA machines are the newest and most expensive lithography tools in the semiconductor industry, and this is their first appearance in a qubit device.

Separately, silicon MOS quantum-dot spin qubits were demonstrated with high fidelity and uniformity using single-exposure 0.33 NA extreme ultraviolet lithography in a 300-millimetre flow [7].

Current State

At the IEDM conference in December 2025, imec used single-print 0.33 NA extreme ultraviolet lithography to pattern all three gate layers of its overlapping-gate quantum-dot architecture [9]. It was the first time that had been done on 300-millimetre wafers. First measurements of those devices showed a room-temperature yield of 90% across the wafer, with back-end functionality confirmed on dedicated test structures [9].

A 90% wafer yield is the kind of number that separates this modality from the rest of this collection.

The change in method explains the result. Switching to single-patterning extreme ultraviolet lithography, replacing a slower electron-beam writer previously used for the three gate layers, gives higher overlay accuracy and better qubit quality [9]. It also allows full 300-millimetre wafers to be printed with high yield and process control.

imec's 300-millimetre flow, using isotopically enriched silicon, produced the Diraq two-qubit devices described in Section 2, and is credited as the low-noise environment behind those fidelities [5]. In a separate collaboration, KIT and imec fabricated spin qubits in a natural, non-enriched silicon-germanium heterostructure, processed entirely on an industrial 300-millimetre line from crystal growth through micromagnet integration [10].

4. Quantum Motion

Quantum Motion built the first complete quantum computer out of standard silicon CMOS and installed it in a national facility. Its architecture is designed to be copied across a chip rather than assembled qubit by qubit.

Recent Developments

Quantum Motion demonstrated a fast spin-qubit readout method reaching 8-microsecond performance with signal amplification, in a 300-millimetre industrial silicon process [11]. Readout speed limits how quickly error correction can run, which connects this result to Part 6.

Current State

The company delivered the industry's first full-stack quantum computer built with standard 300-millimetre silicon CMOS fabrication, installed at the UK National Quantum Computing Centre in September 2025 [12]. The system pairs a silicon spin-qubit processor with a control stack compatible with Qiskit and Cirq, in a three-rack, data-centre-friendly footprint designed for upgrades [12].

The processor uses a tile architecture that integrates compute, readout and control into a dense array which can be printed repeatedly onto a chip [13]. The intent is expansion toward millions of qubits per processor. Repeating a proven tile is the same scaling logic the Fujitsu and RIKEN unit cell follows in Part 1.

Ahead of that delivery, Quantum Motion reported validating a 1,024-quantum-dot array in under five minutes [6]. Speed of characterization matters when a chip holds thousands of dots.

5. Germanium Spin Qubits

Germanium is the main alternative to silicon in this modality. Its qubits are faster and easier to control, and its manufacturing is markedly less mature, so it trades industrial readiness for physics.

Recent Developments

Groove Quantum demonstrated simultaneous operation of an 18-qubit modular array in germanium, built on an extendable architecture [14]. It reported a single-qubit error rate of around 0.2% in a preprint, ahead of peer review [17].

QuTech demonstrated a planar 10-qubit germanium processor in a 3-4-3 layout on a germanium heterostructure, with uniform high-fidelity control [15]. The layout is two-dimensional, so central qubits connect to four neighbours rather than two. Two-dimensional connectivity is what the error-correcting codes in Part 6 assume.

Current State

QuTech's germanium hole-spin work has produced two-qubit gate fidelities above 99.5%, cited as the benchmark for this platform [16].

Groove Quantum is a 2024 spinout from QuTech in Delft, building germanium quantum-dot qubits [17]. The germanium approach offers stronger spin-orbit coupling and faster gate operations than silicon [6]. Its manufacturing process is less mature than the enriched-silicon CMOS path that Intel, Diraq, Quobly and Quantum Motion follow.

One announced figure should not be read as a result. QuTech researchers developed a 23×23 tile chip architecture on a germanium heterostructure, with a stated design capacity to host up to 1,058 hole-spin qubits [18]. That is an architecture's capacity on paper, not a demonstrated qubit count or a fidelity measurement.

6. Donor Qubits at UNSW

Silicon Quantum Computing takes a different approach from every other group in this part. Rather than trapping an electron in a fabricated dot, it places individual phosphorus atoms into silicon with atomic precision.

Current State

The company published a Nature paper in December 2025 demonstrating an 11-qubit atom processor [19]. It uses precision-placed phosphorus donor atoms in isotopically purified silicon, arranged into two multi-nuclear spin registers linked by electron exchange. The processor achieved single-qubit fidelities above 99% and two-qubit gate fidelities up to 99.9% [19].

Unresolved Conflicts

Two figures circulate for the same December 2025 result and they do not agree. The primary report gives two-qubit gate fidelities up to 99.9% for the 11-qubit processor [19]. A secondary source reports Silicon Quantum Computing reaching 99.99% two-qubit fidelity in December 2025, matching a record previously set by trapped-ion platforms [6]. The difference is a factor of ten in error rate, which is the difference between competitive and record-setting. The primary Nature-derived report is the stronger source, but the discrepancy is recorded rather than resolved.

The company restating the figure with its measurement method would settle it.

7. RIKEN

RIKEN holds the highest single-qubit fidelities in this part. Its recent work addresses a problem that appears only once qubits are packed together, which is operating many of them at once without interference.

Recent Developments

RIKEN demonstrated simultaneous high-fidelity single-qubit gates across a spin-qubit array [20]. Primitive $\pi/2$ gate fidelities were well above 99.99% for each qubit individually, with some approaching 99.999%. The method used tailored control pulses on a shared control line together with pairwise phase calibrations.

Sharing one control line across many qubits is how this modality avoids needing one wire per qubit. Doing it without losing fidelity is the result.

Current State

The work uses a silicon-germanium heterostructure with isotopically enriched silicon, and demonstrates simultaneous operation across the array rather than one qubit at a time [20].

RIKEN's Tarucha group separately reported a 5-qubit silicon system with a single-qubit error rate below 0.01%, in a preprint that has not been peer-reviewed [17].

About This Report

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